Four mystery organisms.
Which ones are alive?



Mystery 1



Mystery 2



Mystery 3



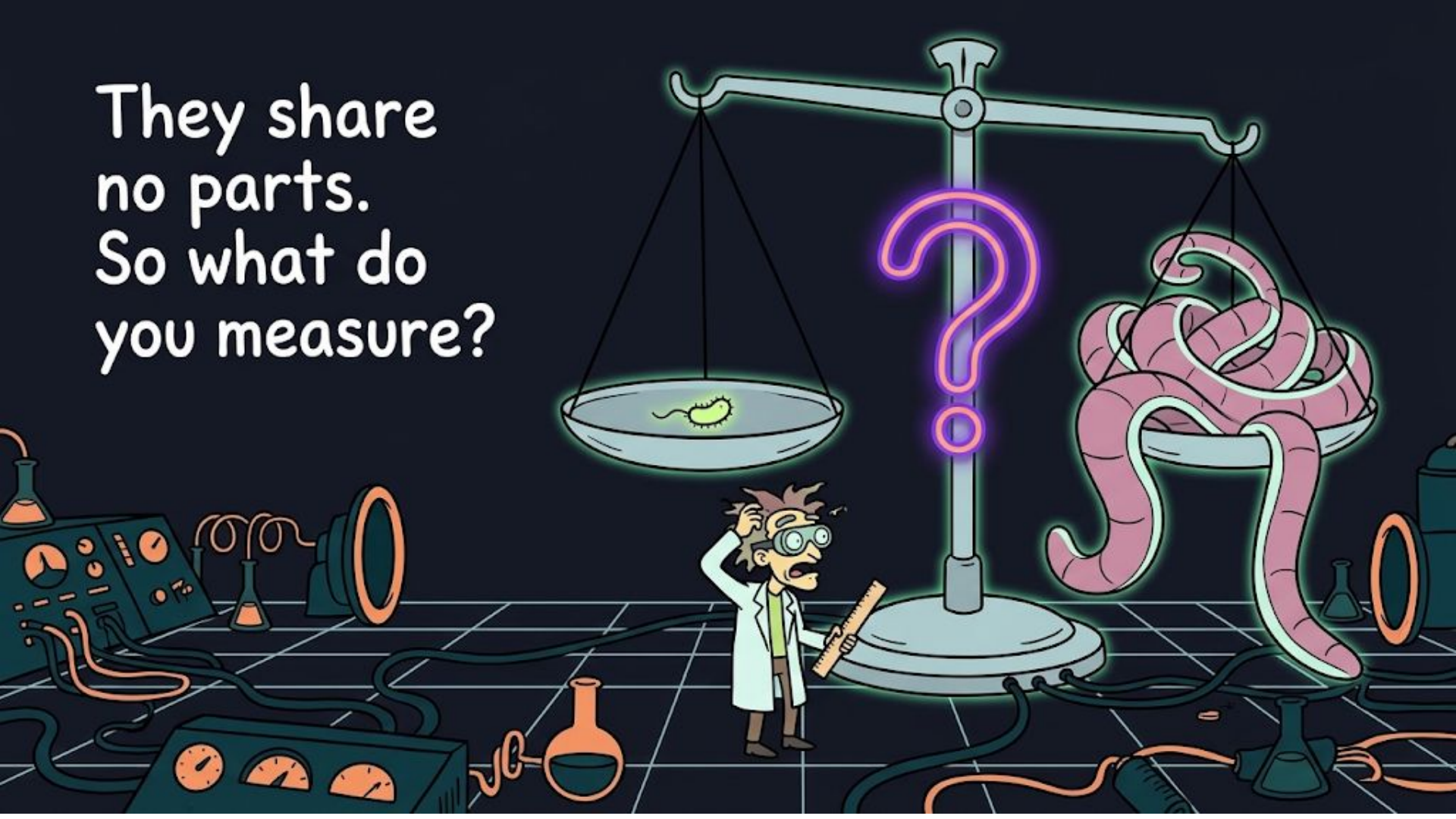

Mystery 4


A bacterium, a virus,
and two parasites.

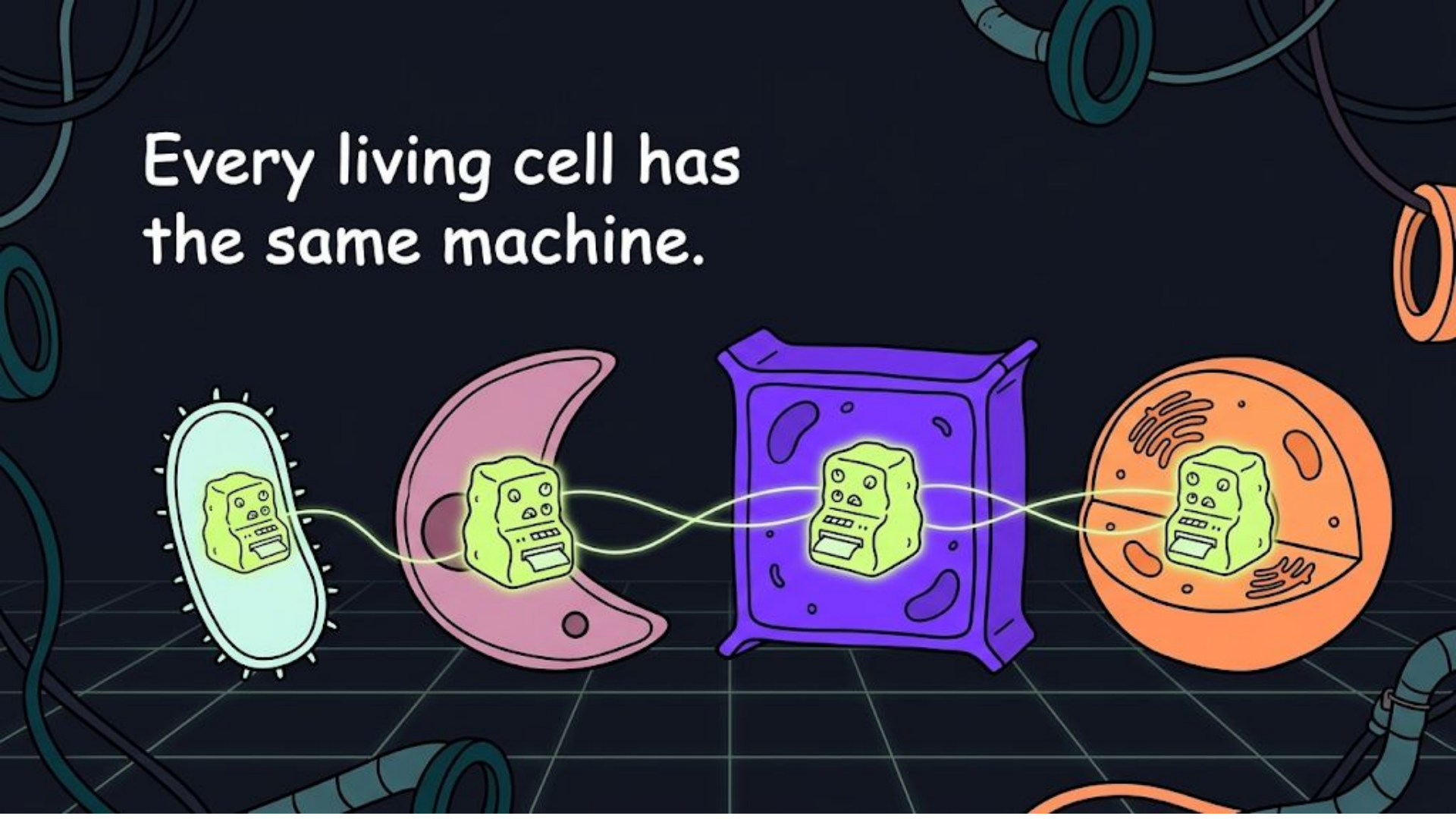

A bacterium, a virus,
and two parasites.

Parasite is a job,
not a family.




They share
no parts.
So what do
you measure?

Every living cell has
the same machine.

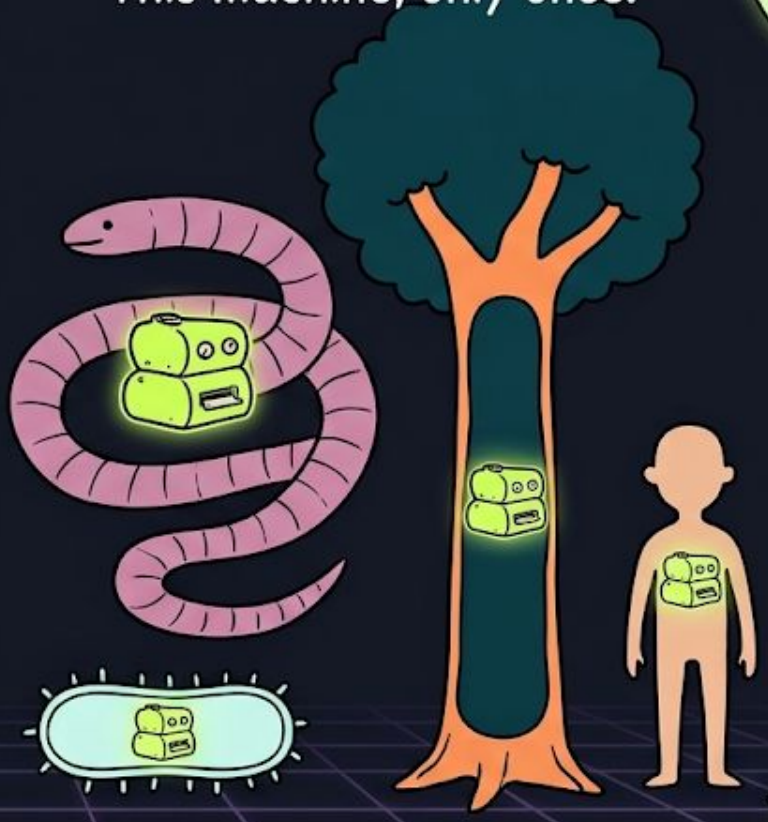


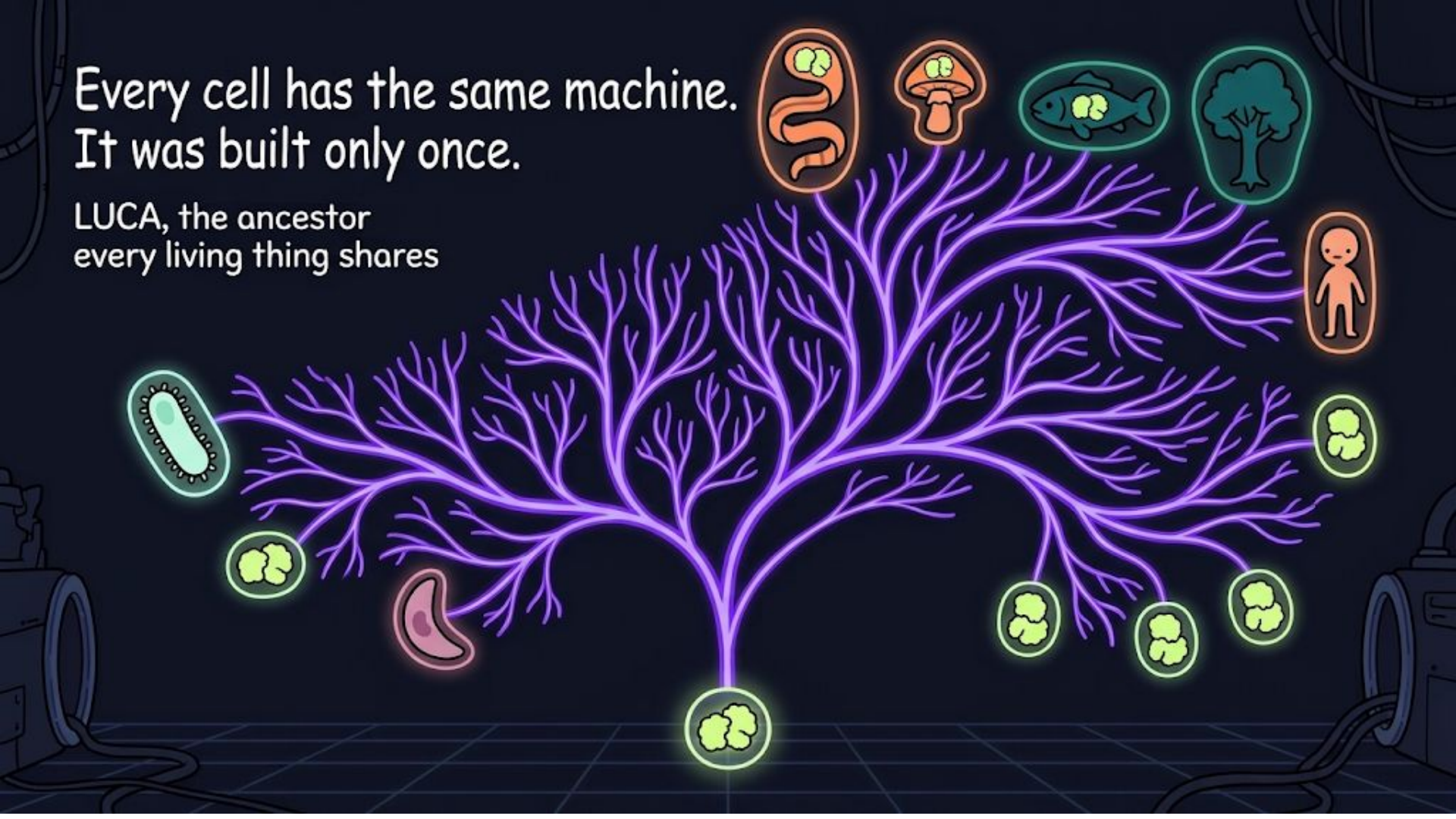
Wings were invented many times.

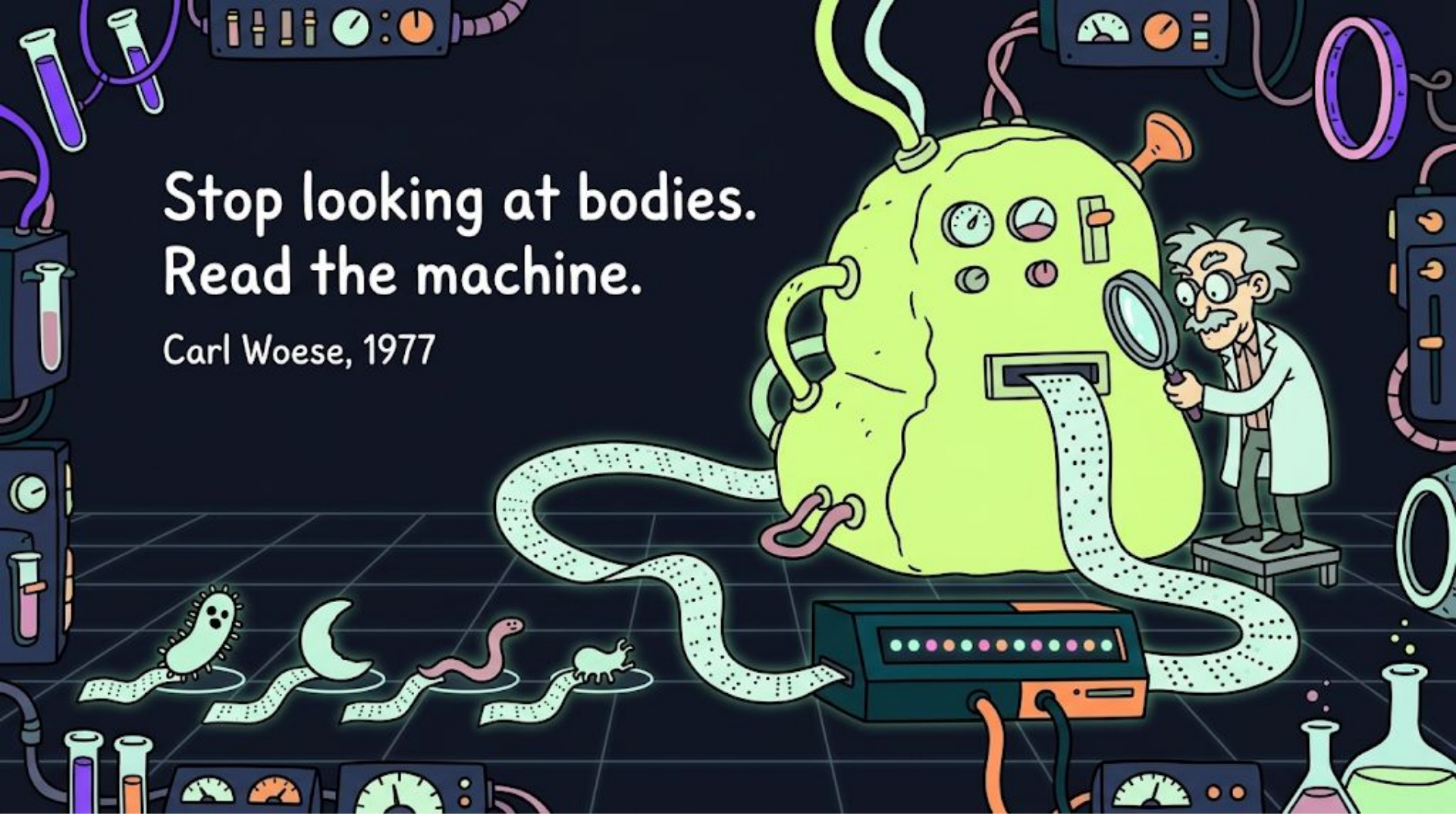
Separate inventions never come out identical.



This machine, only once.




Every cell has the same machine.
It was built only once.
LUCA, the ancestor
every living thing shares


Stop looking at bodies.
Read the machine.
Carl Woese, 1977


No machine, no place on the tree.

BUILD A TREE · 7 MINUTES

Four organisms. Who is closest to whom?

Twenty letters each. No software — just count.

1	Cholera bacterium	GCTAACGAGCAGTGCAATCG
2	Seawater cousin	GCTAACGAGCAGTGTGATCG
3	Gut bacterium	GCTAACGAGTCAGCTAATCG
4	Malaria parasite	TACGTGATTCAGGCTAATCG

	1	2	3	4
1	—			
2		—		
3			—	
4				—

1. Count the differences between every pair. There are 6 pairs.
2. The smallest number is the closest pair.
3. The one with the biggest numbers to everybody is the odd one out.

BUILD A TREE · KEY

Closest pair: 1 and 2. Odd one out: 4.

Highlighted letters differ from sequence 1.

1	Cholera bacterium	GCTAACGAGCAGTGCAATCG
2	Seawater cousin	GCTAACGAGCAGTG**TG**ATCG
3	Gut bacterium	GCTAACGAG**TCAGCT**AATCG
4	Malaria parasite	**TACGTGATT**CAG**GCT**AATCG

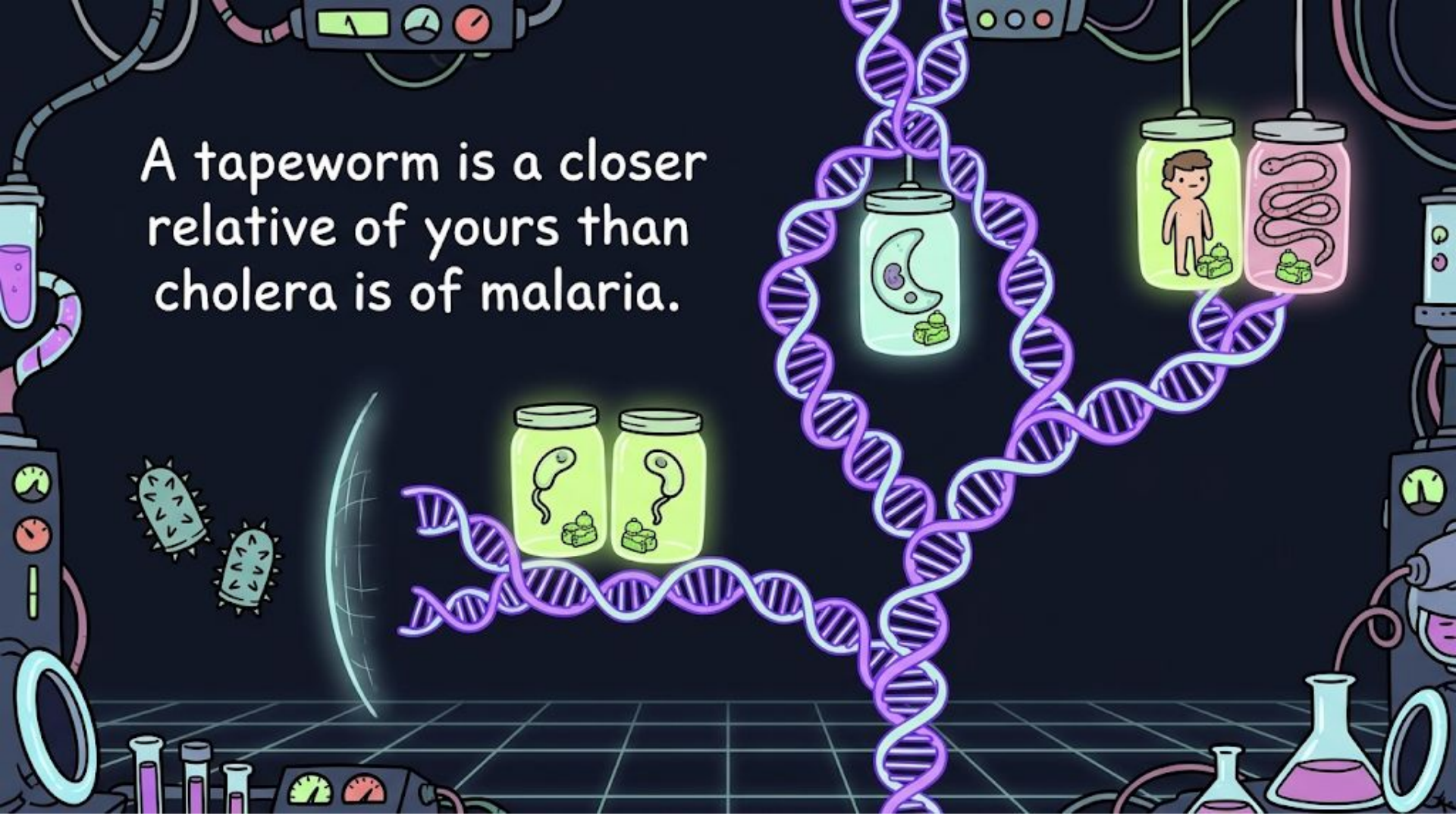
	1	2	3	4
1	—	2	6	12
2	2	—	6	12
3	6	6	—	12
4	12	12	12	—

Number 4 is **12 away from all three** — the same distance to each, and the biggest in the table. Three of these are bacteria. Number 4 is not: it is the malaria parasite, and it has a nucleus.

THE TREE

((1, 2), 3), 4

LIFE SCIENCES · 11–14


A tapeworm is a closer relative of yours than cholera is of malaria.